



Android update bricks Nike shoes

Users bought the Nike's self-lacing Adapt BB sneakers for \$350 a pair only to get them bricked later when updated over Android <https://digit.in/mar19-30>



Twitter test redesigned replies

Twitter is testing redesigned conversations and will accept applications from anyone who wants to try them. <https://digit.in/mar19-31>



Thermally challenged

Agent001 helps sort out a shoddy maintenance job

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A

friend of mine had bought a gaming laptop about a year ago and was very happy with it until

recently. Games were starting to stutter within a few minutes of launch. The graphics weren't cranked up to ridiculous levels, and the only recent software change was GPU driver updates. Obviously, we first downgraded the drivers. No luck. All games were stuttering, which meant it had to be a hardware problem.

After a quick download and installation of some diagnostic software, we were immediately shown bright red warnings for the CPU temperature! It's got to be the CPU fan that's choked with dirt, right? Wrong.

We found that one of the edges of the heatsink mount was elevated. Turns out that whoever did the "maintenance" used a thick thermal pad and didn't seat the heatsink all the way down with the correct pressure. Thermal pads are generally worse for thermal conductivity than thermal pastes, and this one in particular, was pretty hard. It's as if the pad was very old and had already caked up. So we started hunting for thermal pastes that we could use for the CPU. There are a lot of well-known brands that sell thermal pastes and we soon figured that most of them ended up being just about the same.

Most of the difference in cooling is brought about by the heatsink and the attached fans. The thermal paste can bring about a difference of 5-6 degrees Celsius. We are, of course, talking about known brands and not those off-brand containers filled with toothpaste. At the end of the day, the choice of thermal paste comes down to a few simple parameters – thermal conductivity, viscosity, density and the ingredients. Most brands will mention most of these numbers on the packaging but the ingredients are often kept secret. Thermal pastes consist of a base material in which they add materials with higher thermal conductivity. Sometimes they can be electrically conductive as well, which is a good thing because thermal and electric conductivity often go hand-in-hand. However, if the thermal paste were to come in contact with any part of the circuit then we're looking at a nasty short circuit. Imagine if your CPU power pins were to be shorted with the ground pins. If the motherboard does survive, cleaning up the LGA pad is a royal pain. Ideally, you want

something that is thermally conductive, electrically non-conductive and the viscosity should be appropriate for the amount of pressure between the CPU and the heatsink. Also, the compound should last for a year or two without drying up. If the

viscosity is too low, then the thermal paste will slip out from the sides. And if the viscosity is too high, then

you'll have an uneven spread leading to low heat transfer. What we realised was that Zinc Oxide based thermal pastes tend to check all the boxes for the average user. Of which, we ended up shortlisting the Cooler Master Mastergel Maker, Corsair TM30 and Noctua NT-H1. A quick surf on Amazon and we found the Mastergel Maker for ₹499 and went with it. Two days later, we pulled the heatsink apart, cleaned up the existing gunk and lathered on a new layer of thermal grease. Peak temperatures have now dropped to 77 degrees celsius, which is way better than what it was earlier. The thermal paste might have helped by 3-4 degrees in this case, but seating the heatsink the proper way helped the most. 

